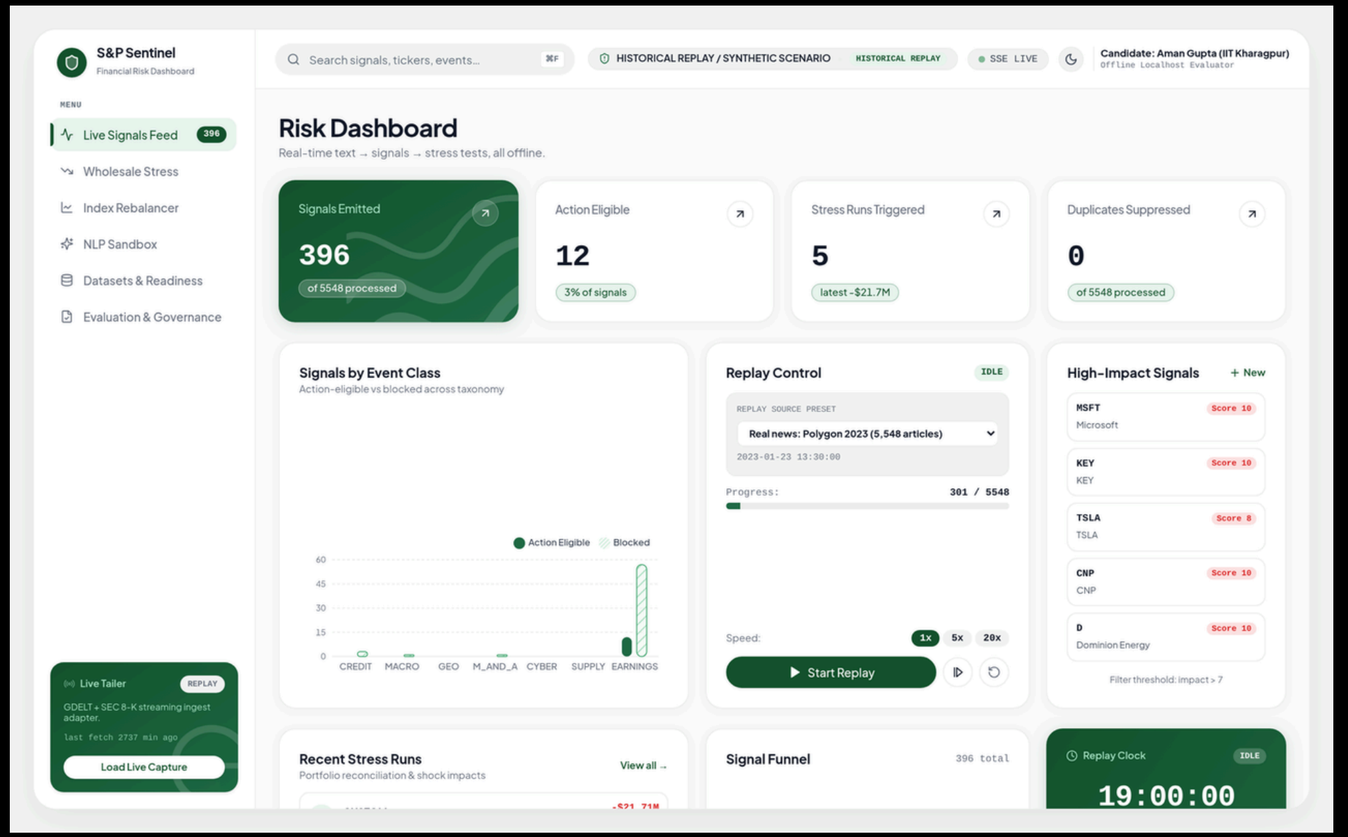
S&P GLOBAL & CRISIL
CAMPUS HACKATHON 2026

S&P Sentinel

Turning scattered risk mentions
into structured signals that drive impact.



Aman Gupta

IIT Kharagpur | Individual Submission

Data: public sources only (Polygon news, Kaggle tweets, GDELT, SEC EDGAR, Yahoo Finance, FRED). No S&P Global or CRISIL data.

Risk events surface first in unstructured text; Sentinel converts every company mention into a structured signal that two downstream modules act on

UNSTRUCTURED TEXT IS NOISY

Critical risk events are buried across news, filings, and reports

Mentions are fragmented, ambiguous, and context-heavy

Traditional feeds surface signals too late

→

ONE SIGNAL PER COMPANY MENTION

Detect and normalize every company mention

Extract risk type, context, and qualifiers

Output a structured signal linked to the company

→

TWO DOWNSTREAM MODULES ACT ON IT

Module A: Index rebalancer

Sentiment re-weights 20 S&P 100 names

Module B: Stress testing

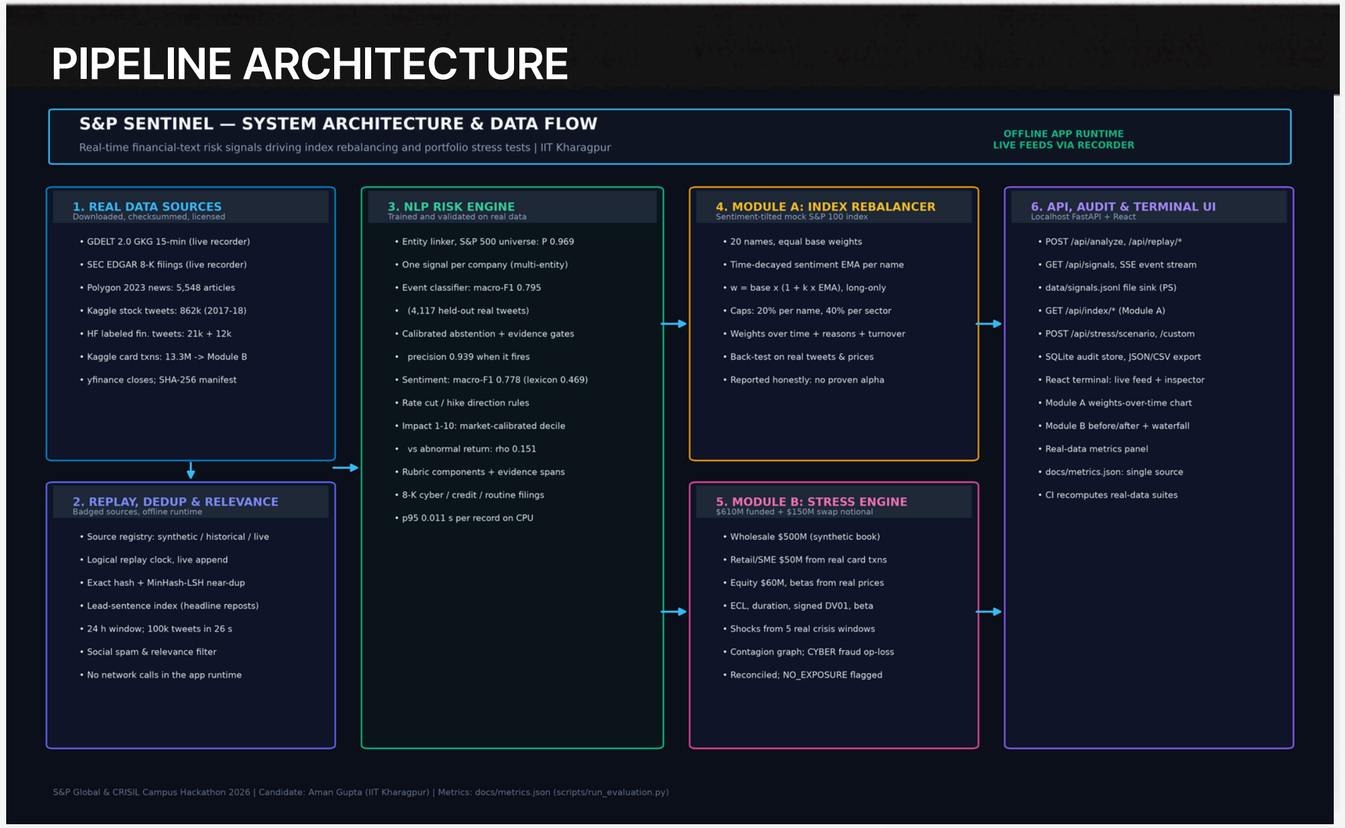
Event-triggered shocks on a $610M book

ANSWER FIRST | **Sentinel turns scattered mentions into structured signals that drive index rebalancing and portfolio stress testing.**

Source: public data only

A single offline pipeline links, classifies, scores and gates every mention in 12 ms at p95 on a laptop CPU

PIPELINE ARCHITECTURE



LATENCY

0.012 s

p95 end-to-end latency on a laptop CPU

NETWORK

0

network calls at runtime

INTERFACES

3

consumption interfaces

Offline processing delivers normalized, classified, scored, and gated signals ready for downstream decisions.

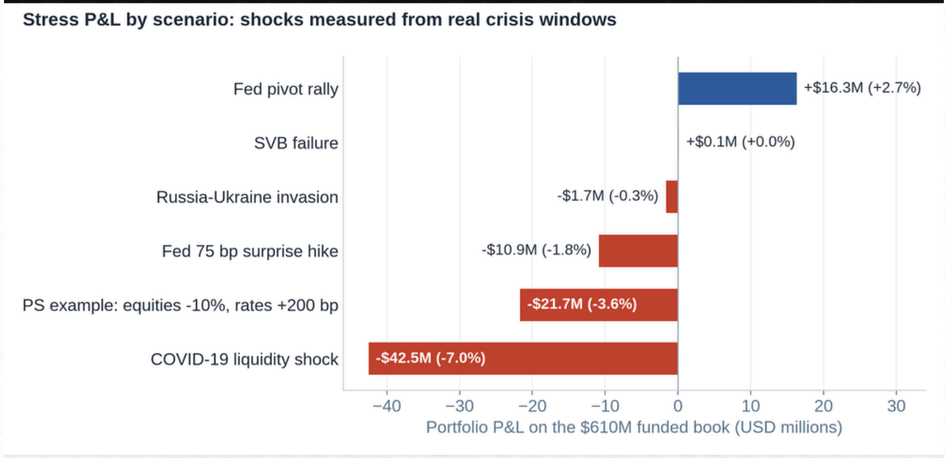
Each component favours auditability over opacity and is benchmarked against a simpler alternative before it ships

COMPONENT	WHAT IT DOES	WHY THIS CHOICE
Entity linker	Normalizes mentions to the right company	Rule-based, auditable: P 0.969 / R 0.981
Event classifier	Detects and labels risk events	TF-IDF + calibrated logistic regression
Sentiment	Measures tone and direction	Real-data model; beat FinBERT prototype
Impact 1-10	Scores financial impact on a 1-10 scale	Boosted model of next-day abnormal return
Dedup + relevance	Removes duplicates and filters weak hits	Exact hash + MinHash-LSH, 24 h window
Stress engine	Propagates signals through stress frameworks	ECL, duration, DV01, beta; reconciles

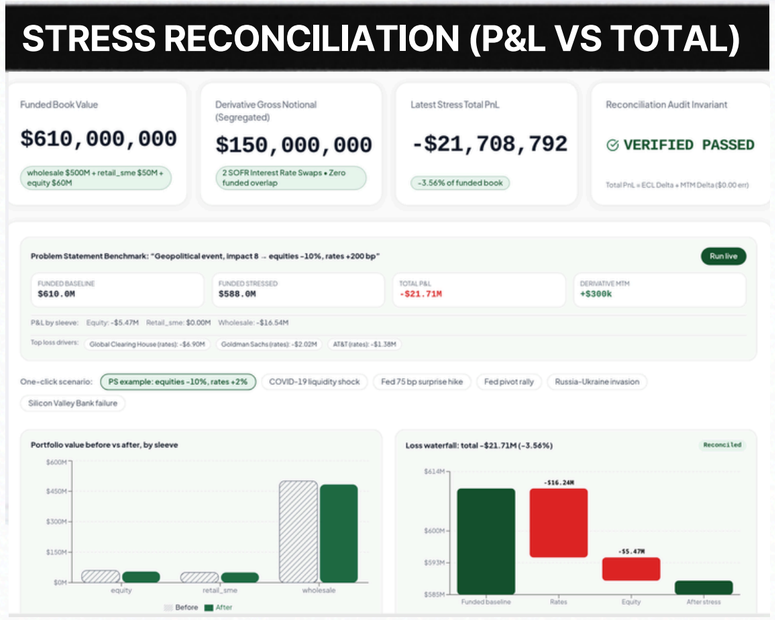
Models beat naive baselines on real held-out data, and every stress run reconciles position losses to the total

Event macro-F1	Sentiment macro-F1	Entity linking precision	Market lift
0.795	0.778	0.969	2.64x
vs 0.286 keyword baseline (4,117 held-out tweets)	vs 0.469 lexicon baseline (2,388 held-out tweets)	recall 0.981 on 2,521 real news articles	eligible signals vs base rate, Q4 2023 holdout

SCENARIO P&L ON THE $610M FUNDED BOOK



STRESS RECONCILIATION (P&L VS TOTAL)



Module A back-test (443 days): active return +0.02%/yr, not significant; no proven alpha.

Calibrating shocks on real crises avoids the textbook error that every crisis lifts interest rates

FLIGHT TO QUALITY DYNAMICS

Crisis episodes often compress yields as investors move to safety

FRED SPREAD VALIDATION

Use real historical spreads to ground shock direction and magnitude

REAL EXPOSURES, SYNTHETIC SIZES

Map actual portfolio exposures to economic shocks and scenarios

DEFENSIBLE ANALYST DECISIONS

Evidence-based assumptions that withstand review and challenge

SO WHAT | **Real-world calibration reveals counterintuitive moves, strengthening scenario P&L and stress testing.**

The prototype states its gaps openly; four next steps would take it towards production

LIMITATIONS

- AI-labelled news holdout, not yet reviewed
- Tweet-trained classifier abstains on much real news
- 0 automatic stress triggers on Q4 2023 holdout
- Simplified stress model, synthetic positions

NEXT STEPS

1. Human-review the news holdout labels
2. Improve models for accuracy and robustness
3. Add user feedback to refine signal quality
4. Integrate with enterprise risk and workflow systems

BOTTOM LINE | **Openly acknowledging limitations guides focused execution towards a production-ready S&P Sentinel.**